Technical Data
Data Sheet N0623, Rev. A

MBR20100CT SCHOTTKY RECTIFIER

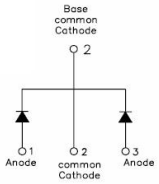


TO-220AB

Features

- 150˚C T_J operation
- Center tap configuration
- Low forward voltage drop
- High purity, high temperature epoxy encapsulation for enhanced
- mechanical strength and moisture resistance
- High frequency operation
- Guard ring for enhanced ruggedness and long term reliability
- This is a Pb – Free Device
- "-HF" suffix is for Halogen Free Device
- All SMC parts are traceable to the wafer lot
- Additional testing can be offered upon request

Circuit Diagram



Applications

- Switching power supply
- Converters
- Free-Wheeling diodes
- Reverse battery protection

Maximum Ratings:

Characteristics	Symbol	Condition	Max.	Units
Peak Repetitive Reverse Voltage Working Peak Reverse Voltage DC Blocking Voltage	V_{RRM} V_{RWM} V_R	-	100	V
Average Rectified Forward Current	$I_{F(AV)}$	50% duty cycle @Tc=105˚C, rectangular wave form	10(Per Leg) 20(Per Device)	A
Peak One Cycle Non-Repetitive Surge Current(Per Leg)	I_{FSM}	8.3ms, Half Sine pulse, T_C = 25 ˚C	150	A

Electrical Characteristics:

Characteristics	Symbol	Condition	Typ.	Max.	Units
Forward Voltage Drop(Per Leg)*	V_{F1}	@ 10A, Pulse, T_J = 25 ˚C	0.75	0.90	V
	V_{F2}	@ 10A, Pulse, T_J = 125 ˚C	0.63	0.80	V
Reverse Current(Per Leg)*	I_{R1}	@V_R = rated V_R, T_J = 25 ˚C	0.006	1.0	mA
	I_{R2}	@V_R = rated V_R, T_J = 125 ˚C	1	6.0	mA
Junction Capacitance(Per Leg)	C_T	@V_R = 5V, T_C = 25 ˚C f_{SIG} = 1MHz	230	300	pF
Voltage Rate of Change	dv/dt	-	-	10,000	V/μs

* Pulse width < 300 μs, duty cycle < 2%

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Thermal-Mechanical Specifications:

Characteristics	Symbol	Condition	Specification	Units
Junction Temperature	T_J	-	-55 to +150	˚C
Storage Temperature	T_{stg}	-	-55 to +150	˚C
Typical Thermal Resistance Junction to Case	$R_{\theta JC}$	DC operation	3.5	˚C/W
Approximate Weight	wt	-	2	g

Ratings and Characteristics Curves

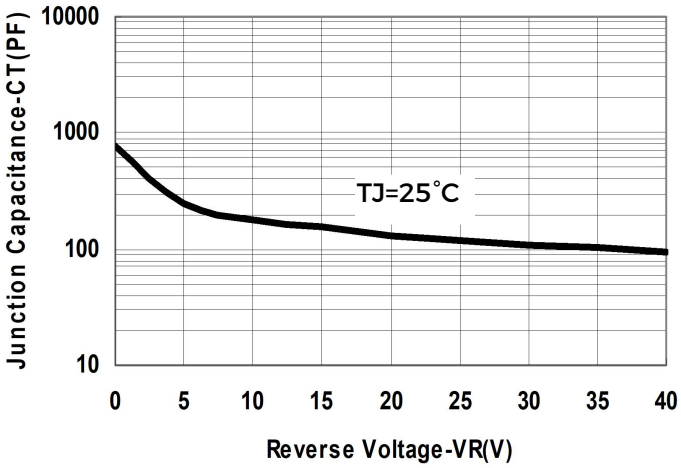


Fig.1-Typical Junction Capacitance

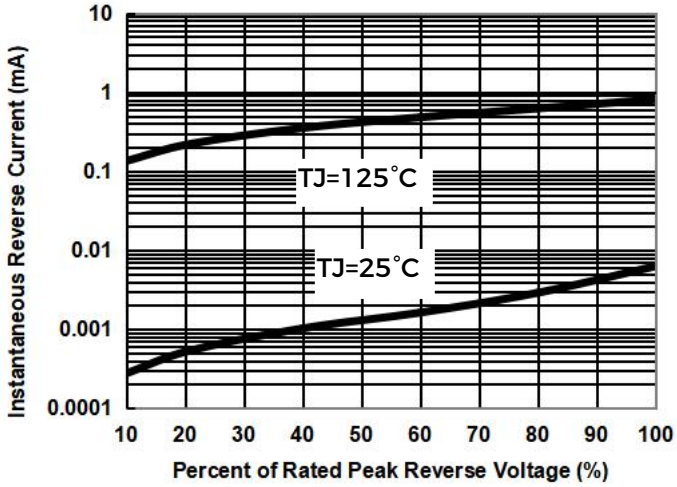


Fig.2-Typical Reverse Characteristics

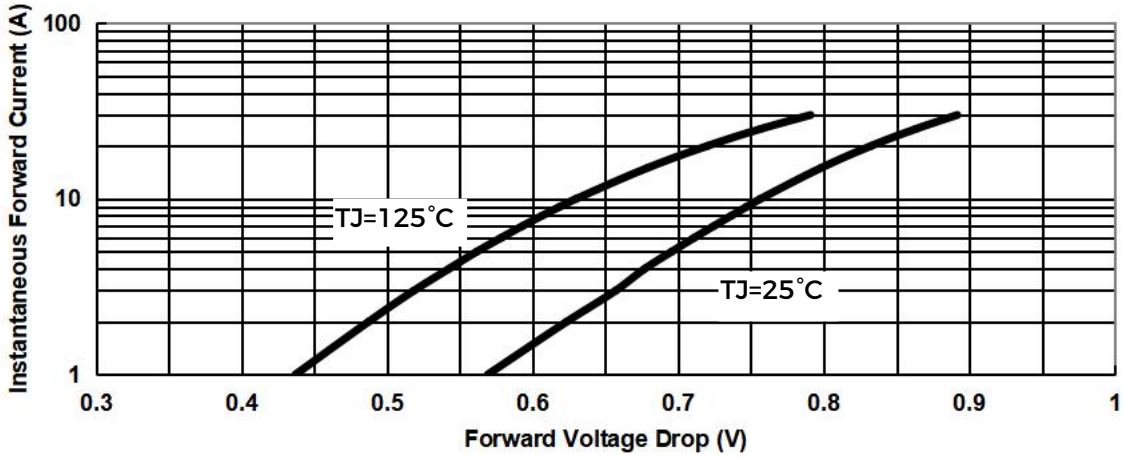
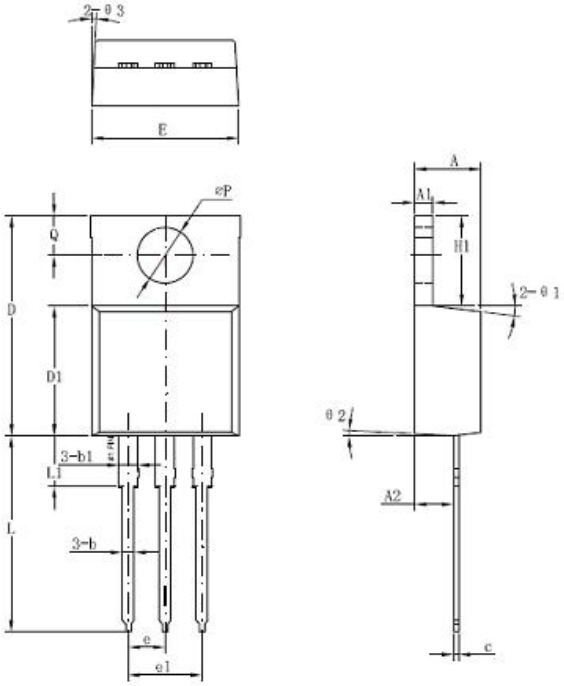


Fig.3-Typical Instantaneous Forward Voltage Characteristics

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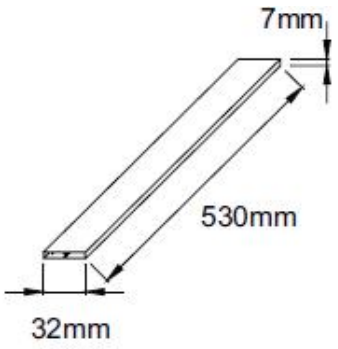
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Mechanical Dimensions TO-220AB

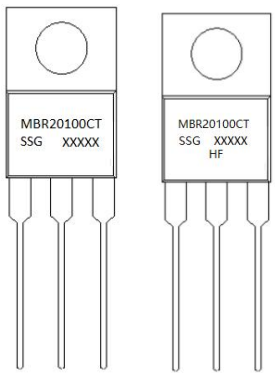


Symbol	Dimensions in millimeters		
	Min	Typical	Max
A	4.42	4.57	4.72
A1	1.17	1.27	1.37
A2	2.52	2.69	2.89
b	0.71	0.81	0.96
b1	1.17	1.27	1.37
c	0.31	0.38	0.61
D	14.94	15.24	15.54
D1	8.85	9.00	9.15
E	10.01	10.16	10.31
e		2.54	
e1	4.98	5.06	5.18
H1	6.04	6.24	6.44
L	12.7	13.56	13.80
L1	3.56	3.5	3.96
ΦP	3.74	3.84	4.04
Q	2.54	2.74	2.94
θ1		7°	
θ2		3°	
θ3		4°	

Tube Specification



Marking Diagram



MBR20100C MBR20100CT-HF

Where XXXXX is YYWWL

MBR	= Device Type
20	= Forward Current (20A)
100	= Reverse Voltage(100V)
CT	= Configuration
HF	= Halogen Free
SSG	= SSG
YY	= Year
WW	= Week
L	= Lot Number

Cautions: Molding resin
Epoxy resin UL:94V-0

Ordering Information

Device	Package	Shipping
MBR20100CT	TO-220AB (Pb-Free)	50 pcs/ tube

For information on tape and reel specifications, including part orientation and tape sizes, please refer to our tape and reel packaging specification.

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